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Strategic Allocation of Battery Energy Storage Systems in Multi-Service Markets: An EPEC Modeling Approach

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Abstract

This thesis aims to determine optimal allocation strategies—specifically siting and sizing in MW and MWh—for BESS in multi-service markets, bridging the gap between technical grid constraints and economic profitability. To accurately capture the oligopolistic behavior of strategic investors, an Equilibrium Problem with Equilibrium Constraints (EPEC) framework is developed. The upper level models independent investors maximizing their individual Net Present Value (NPV), while the lower level represents an Independent System Operator (ISO) minimizing system costs via a direct current optimal power flow (DC-OPF) and an automatic Frequency Restoration Reserve (aFRR) reserve market clearing. The resulting non-linear programming (NLP) problems are solved utilizing a diagonalization algorithm. The results demonstrate that investors aggressively distribute their capacities across the entire network to exhaust all available connection limits and maximize their market shares. A major finding of this study is a severe Arbitrage-Reserve Trade-off, demonstrating that lucrative capacity premiums in the aFRR market override traditional spot market logic. The thesis concludes that evaluating technical restrictions and multi-market bidding in isolation distorts BESS investment cases. Future research should focus on integrating co-located hybrid renewable park configurations.

Contents

Abstract	iii
1 Introduction	1
1.1 Motivation and Background	1
1.2 Core Objectives and Research Question	1
1.3 Method Applied	2
1.4 Outline of the Master Thesis	2
2 State-of-the-Art and Progress Beyond	3
2.1 State-of-the-Art: BESS Allocation in Power Grids	3
2.2 State-of-the-Art: EPECs in Energy Systems Modeling	4
2.3 Contribution	5
3 Methodology	7
3.1 Equilibrium Problem with Equilibrium Constraints	7
3.2 Upper Level: The Strategic Investor	10
3.3 Lower Level: Market Clearing	11
3.4 Model Expansions: Bridging the Research Gap	12
3.4.1 Battery Degradation	12
3.4.2 Co-Optimization of Spot and aFRR Markets	13
3.4.3 Economic RES curtailment	14
3.4.4 Discarded model features	15
3.5 EPEC Solution Strategy and Algorithm	15
3.5.1 The Diagonalization Algorithm	15
3.5.2 Convergence Criteria and Multiple Equilibria	16
3.5.3 Sequential vs. Parallel Execution	16
3.5.4 Technical Tweaks for Computational Speed	17
3.5.5 Model Tweaks for Stability and Tractability	17
3.6 Input Parameters and Empirical Scale	19
3.6.1 Grid Topology and Generation Fleet	20
3.6.2 Demand and Renewable Energy Profiles	21
3.6.3 BESS Characteristics and Costs	21

Contents

4	Results & Discussion	23
4.1	Algorithmic Performance and EPEC Stability	23
4.2	Strategic BESS Allocation: Siting, Sizing, and the E/P Ratio	25
4.3	Revenue Stream Breakdown and the Arbitrage-Reserve Trade-off	26
4.4	System-Wide Operation Profiles and Grid Congestion	28
4.5	Nodal Price Dynamics and Local Arbitrage	29
5	Synthesis of the Results	35
5.1	Answers to the Research Question	35
5.2	Strengths and Limitations of the EPEC Methodology	36
5.2.1	Strengths of the Applied Framework	36
5.2.2	Limitations and Mathematical Compromises	37
5.3	Upscaling and Transferability of Results	38
5.3.1	Temporal and Spatial Upscaling	38
5.3.2	Transferability to European Market Designs	39
5.3.3	Algorithmic Evolution towards MIPECs	39
5.3.4	Transferability to Co-located Hybrid Configurations	40
5.4	Lessons Learned from EPEC Modeling	40
5.4.1	Algorithmic Limitations in Capturing Economic Rationality	40
5.4.2	Challenges in Computing Simultaneous Market Equilibria	41
6	Conclusions and Outlook	43
	Bibliography	45

1 Introduction

1.1 Motivation and Background

The ongoing shift from fossil fuels to renewable energy sources (RES) brings new challenges with it. Weather dependent RES like solar or wind energy have high generation variability, which has a negative impact on the power system, creating reliability on the controllable, traditional power generation by fossils. To counteract the negatives, stabilize the generation output and therefore complete the energy transition, stakeholders are more and more implementing energy storage systems (ESS) projects into the power grid. One of the fastest acting energy storage technologies (EST) with high volumetric energy density, high power density and round-trip efficiency are battery energy storage systems (BESS) like Lithium-ion batteries. For these reasons and due to favorable price-development, BESS make the second biggest operational EST capacity globally, following pumped hydro [1]. With fast growing BESS capacity, there is high potential for optimization in terms of sizing and siting future BESS projects in order to reach higher technical and economic efficiency and finally speed up the energy transition.

1.2 Core Objectives and Research Question

This thesis aims to develop an optimization-based modeling framework to determine optimal allocation strategies for Battery Energy Storage Systems, considering both technical and economic constraints for transmission grid operators.

The core objective is to determine the optimal generation and storage capacity of BESS – expressed in MW and MWh, respectively – at the nodal grid level.

To guide this analysis and bridge the identified gaps in current energy system modeling, the following exploratory research question is posed:

What insights into strategic investor behavior can be gained by shifting from a traditional central-planner cost-minimization approach to a decentralized, game-theoretic EPEC framework when determining optimal BESS capacities?

1.3 Method Applied

To fulfill these objectives, this thesis departs from traditional central-planner cost-minimization approaches and adopts a game-theoretic perspective. In liberalized electricity markets, investment decisions are made by independent, profit-maximizing actors rather than a single entity, as highlighted by Conejo and Ruiz [2]. Therefore, the interactions of strategic BESS investors are modeled as an Equilibrium Problem with Equilibrium Constraints (EPEC).

An EPEC is a set of interrelated bi-level optimization problems, known as Mathematical Programs with Equilibrium Constraints (MPECs) [2]. In this framework, the upper level represents the strategic BESS investors maximizing their individual Net Present Value (NPV), while the lower level represents the Independent System Operator (ISO) clearing the market via a cost-minimal direct current optimal power flow (DC-OPF). Similar EPEC formulations have been successfully applied in the energy system context to analyze equilibria in coupled electricity and natural gas markets, demonstrating the method's capability to capture complex, strategic market dynamics [3].

To calculate the cost-minimal and profit-maximizing investment decisions, the proposed mathematical optimization model is implemented in Python. The complex mathematical formulation of the non-linear complementarity conditions is built using the Pyomo optimization package [4]. The resulting non-linear programming (NLP) problems are subsequently solved using the state-of-the-art interior-point optimizer Ipopt [5], utilizing a diagonalization algorithm to compute the Nash Equilibria among the investors.

1.4 Outline of the Master Thesis

This work is organized as follows. Chapter 2 provides a comprehensive state-of-the-art review of BESS allocation in power grids and the application of EPECs in energy system modeling, defining the novelty and progress beyond the current literature. Chapter 3 elaborates on the applied EPEC methodology, detailing the mathematical formulation, the model expansions and the algorithmic solution strategy. Chapter 4 presents the quantitative results and discusses the strategic investment behavior, grid congestion, and the "Negative Arbitrage" phenomenon. Chapter 5 synthesizes these findings, answers the core research question, and critically evaluates the strengths and limitations of the EPEC approach. Finally, Chapter 6 concludes the thesis and provides an outlook for future research.

2 State-of-the-Art and Progress Beyond

This chapter summarizes the current state of knowledge in the literature regarding BESS allocation and advanced energy system modeling. Section 2.1 provides a comprehensive review of BESS allocation in power grids, utilizing a multi-dimensional taxonomy to identify the prevailing research gap between strictly technical grid simulations and purely economic optimization approaches. Section 2.2 explores the application of EPECs in modern energy economics, highlighting their capability to realistically capture strategic, oligopolistic market behavior. Finally, Section 2.3 synthesizes these findings to clearly define the genuine contribution of this thesis and outline the progress beyond the current state-of-the-art.

2.1 State-of-the-Art: BESS Allocation in Power Grids

The allocation of BESS in power grids has been extensively studied from various methodological perspectives. By classifying recent literature through the multi-dimensional taxonomy in Table 2.1 – focusing on optimization methods, objective functions, and constraints – distinct trends and research gaps emerge.

Regarding the **optimization methods**, the literature is broadly divided into meta-heuristic approaches and deterministic models. Meta-heuristic algorithms, such as Genetic Algorithms (GA) or Particle Swarm Optimization (PSO), are frequently used to navigate the non-linearities of BESS placement and sizing [7]. Conversely, deterministic approaches utilize Linear Optimal Power Flow (LOPF) to find guaranteed optimal solutions for specific system setups, often minimizing total system costs [15]. Other works employ highly intensive brute-force searches to evaluate complex, non-linear battery degradation models for BESS sizing [8].

When analyzing the **objective functions and constraints**, a significant dichotomy becomes apparent. Most studies focus strictly on either technical or economic goals. Technical models prioritize grid stability, utilizing BESS to minimize frequency deviations, Rate of Change of Frequency (ROCOF), or voltage fluctuations [7, 14]. In these models, economic realities like investment costs are often heavily simplified. On the other hand, purely economic models aim to maximize the Net Present Value (NPV)

2 State-of-the-Art and Progress Beyond

Table 2.1: Taxonomy of BESS allocation in power grids. (Aug.) refers to battery augmentation, whereas (Dist.) refers to distribution across existing generators.

Literature	Sizing	Siting	Focus	Optimization Method	Objective function	Constraints	Time Horizon	Local Focus
[6]	X	X	Both	Hybrid MOPSO & NSGAI	Min. Cost, Voltage Deviation, CO ₂	Grid Simulation	Today	Generic (IEEE 30-bus)
[7]	X		Technical	FSCABC, verified with PSO	Min. Freq. Deviation & ROCOF	Grid Sim (Freq. Nadir, ROCOF)	Today/Near-Future	Generic (IEEE 39-bus)
[8]	X (Aug.)		Economical	Iterative Brute-Force Search	Max. Net Present Value (NPV)	Battery Degradation, Market Sim (PPA)	Future (15-year)	South Korea
[9]	X	X	Technical	Improved PSO (IPSO)	Min. Voltage Deviation & Power Loss (Fuzzy)	Grid Simulation	Today	Generic (IEEE-33 bus)
[10]		X	Economical	Qualitative Scoring Framework	Max. Business Feasibility	External/Business Factors (Land, Permits)	Today/Near-Future	Denmark (Bornholm)
[11]	X	X	Technical	GA with Monte Carlo Sim	Min. BESS Units & Solar Curtailment	Grid Simulation (DC-OPF)	Today	Generic (IEEE 24-bus)
[12]	X		Economical	BPNN (prediction) + GA (opt.)	Min. Investment Cost	Reliability (LPSP), Grid Sim	Today/Near-Future	Kenya (LTWPP)
[13]	X	X	Technical	Lssvm (partition) + ResNet DNN	Max. Energy Throughput	Grid Simulation	Today	Generic (IEEE 33-node)
[14]	X (Dist.)		Technical	Analytical (Swing Equation)	Limit ROCOF per Area	Local Frequency Dynamics	Future	Generic & Cape Verde
[15]		X	Economical	Linear Optimization (LOPF)	Min. Total System Costs	Grid Simulation (DC-OPF)	Future (2030)	Germany

through energy arbitrage, but often neglect strict physical grid constraints (such as nodal congestion) or the physical degradation of the battery assets [8].

Ultimately, the current state-of-the-art lacks a comprehensive, hybrid approach. There is a distinct gap for optimization frameworks that seamlessly combine detailed economic profit maximization (such as multi-service market participation in both spot and reserve markets) with strict technical and physical grid constraints.

2.2 State-of-the-Art: EPECs in Energy Systems Modeling

While traditional cost-minimization models (like central-planner LOPF) are valuable for theoretical grid studies, they fail to represent the reality of liberalized electricity markets, where independent actors maximize their own profits. As emphasized by Conejo and Ruiz, “Complementarity, not Optimization, is the Language of Markets” [2]. To accurately model these strategic interactions, the methodology of Equilibrium Problems with Equilibrium Constraints (EPECs) has gained significant traction.

An EPEC finds the Nash Equilibrium among multiple strategic actors, where each actor's problem is formulated as a Mathematical Program with Equilibrium Constraints (MPEC) [2]. This methodology has proven highly effective in modern energy economics. For instance, Chen et al. successfully applied an EPEC to determine equilibria in coupled electricity and natural gas markets, demonstrating the method's ability to capture complex strategic offering and bidding behaviors [3].

Particularly relevant to this thesis is the work by Wogrin et al., who utilized an EPEC to model generation capacity expansion in liberalized markets [16]. They proved that a bilevel approach – where investors maximize profits in the upper level while anticipating the market clearing in the lower level – yields vastly different and more realistic investment schedules compared to traditional single-stage cost-minimization models. However, while EPECs are established for conventional generation expansion, their application to the highly complex, constraint-heavy allocation of multi-use BESS remains a largely unexplored frontier.

2.3 Contribution

Based on the identified gaps in BESS allocation and the proven strengths of complementarity modeling, this thesis creates a novel methodological bridge. The genuine contribution of this work lies in the development of a comprehensive EPEC framework tailored specifically to the multi-service allocation of BESS.

The progress beyond the state-of-the-art is defined by the following key innovations:

- **Decentralized Oligopoly Modeling:** Shifting away from central-planner assumptions by modeling strategic BESS investors who actively compete for nodal grid capacities and market shares.
- **Simultaneous Siting and Sizing:** Endogenously determining both the optimal location and the exact Energy-to-Power ratio (MW and MWh) based on individual investor risk affinities (WACC).
- **Multi-Service Co-Optimization:** Bridging the gap between technical and economic models by co-optimizing day-ahead spot market arbitrage with the strict physical requirements of continuous reserve market provision (aFRR).
- **Internalized Physical Constraints:** Integrating linearized cyclic battery degradation directly into the strategic objective function, ensuring that investment decisions reflect long-term operational wear.

Based on the identified gaps in BESS allocation and the proven strengths of complementarity modeling, this thesis creates a novel methodological bridge. The genuine contribution of this work lies in the development of a comprehensive EPEC framework

2 State-of-the-Art and Progress Beyond

tailored specifically to the multi-service allocation of BESS. So far, the literature has predominantly focused on either purely technical grid simulations or centralized economic cost-minimization models, while the application of strategic equilibrium models has largely been restricted to conventional generation expansion.

In contrast, this thesis shifts away from traditional central-planner assumptions by introducing a decentralized oligopoly model representing strategic BESS investors who actively compete for nodal grid capacities and market shares. Rather than relying on simplified heuristics, the framework simultaneously determines the optimal location and the exact Energy-to-Power ratio (MW and MWh) based on individual investor risk affinities. Furthermore, it bridges the gap between technical and economic models through a multi-service co-optimization that balances day-ahead spot market arbitrage with the strict physical requirements of continuous reserve market provision (aFRR). Ultimately, this approach fully internalizes physical constraints by integrating linearized cyclic battery degradation directly into the strategic objective function, ensuring that the resulting investment decisions realistically reflect long-term operational wear.

3 Methodology

The core of this research is the development of a modeling framework designed to determine optimal allocation strategies for Battery Energy Storage Systems (BESS). Unlike conventional cost-minimization models that assume a central planner, this work adopts a game-theoretic perspective to capture the strategic interactions between competing private entities. Within this framework, four strategic investors compete across a five-node network for market share and limited grid connection capacity. The optimization is conducted over a representative day with an hourly resolution, accurately internalizing long-term capital expenditures by scaling them to daily costs.

This chapter outlines the core methodology of the thesis, presenting the Equilibrium Problem with Equilibrium Constraints used to model strategic BESS allocation. It sequentially details the underlying Mathematical Programs with Equilibrium Constraints, the integration of technical and economic model expansions, the chosen algorithmic solution strategy, and finally the empirical parameters configuring the test system.

3.1 Equilibrium Problem with Equilibrium Constraints

The mathematical structure employed is an **Equilibrium Problem with Equilibrium Constraints (EPEC)**. An EPEC is fundamentally a "game" played between multiple strategic leaders who all face a common follower [2]. In the context of this thesis, the EPEC represents the simultaneous competition of several BESS investors for market share and grid connection capacity.

As shown in Figure 3.1, the EPEC is composed of several interrelated **Mathematical Programs with Equilibrium Constraints (MPECs)**, where each MPEC represents the decision-making process of a single strategic investor. Because the objective function of each investor depends not only on their own decisions but also on the investment and operational choices of their rivals, these MPECs are intrinsically coupled [2].

Each individual MPEC is structured as a bi-level optimization problem, reflecting a Leader-Follower (Stackelberg) relationship.

3 Methodology

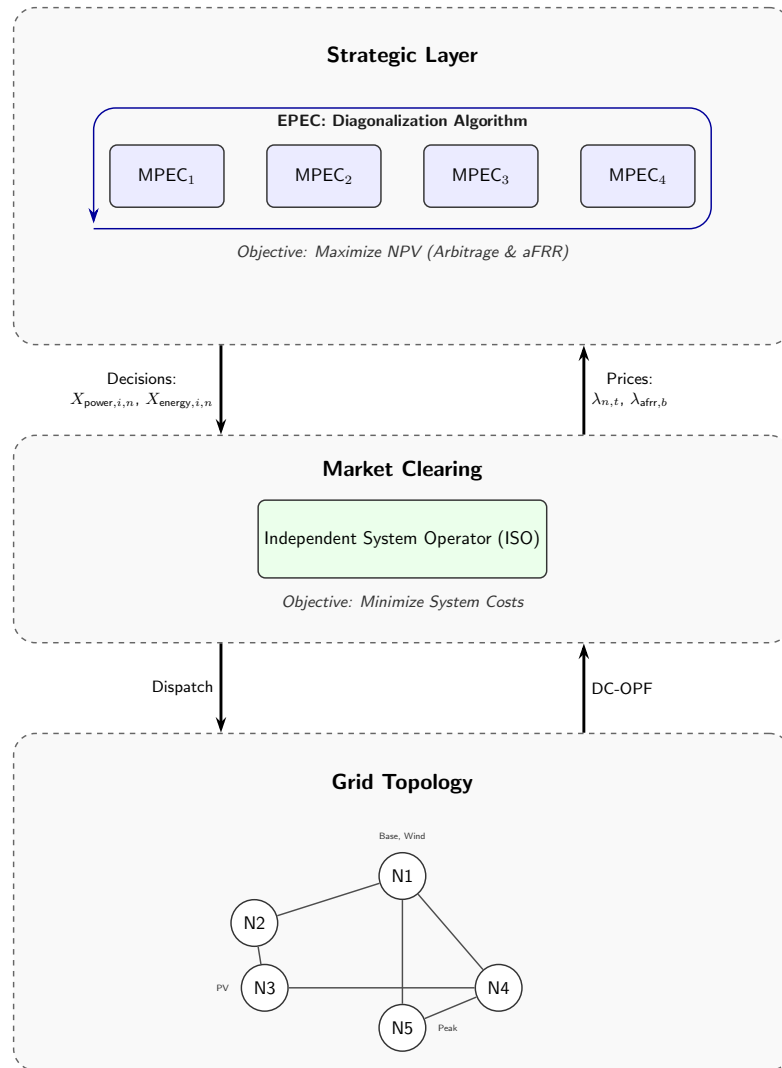


Figure 3.1: Modeling approach of BESS investors as strategic agents (MPEC) coupled in their equilibria (EPEC).

3.1 Equilibrium Problem with Equilibrium Constraints

- **Upper Level (The Leader):** This level represents the strategic BESS investors. Their primary goal is to maximize their individual Net Present Value (NPV). The decision variables at this level are the investment choices: the power capacity (X_{power}) in MW and the energy capacity (X_{energy}) in MWh to be installed at specific nodes within the grid.
- **Lower Level (The Follower):** This level represents the Independent System Operator (ISO) or market clearing mechanism. The ISO performs a cost-minimal DC Optimal Power Flow (DC-OPF), ensuring that the system's power balance and technical grid constraints – such as line limits – are respected while meeting demand.

The levels are coupled through the Locational Marginal Prices (LMPs): the ISO clears the market, and the resulting nodal prices are fed back to the Upper Level, where they determine the revenues and thus the profitability of the investors' BESS assets.

The investors in this model are characterized as strategic agents who act according to the principle of economic rationality. This means each investor assumes that all other market participants (rivals and the ISO) will also act to maximize their own objectives. To ensure this behavior is mathematically captured, the Lower Level market clearing is replaced by its KKT optimality conditions. These conditions are then embedded as constraints within the investors' Upper Level optimization, forcing the strategic actors to internalize the market's response to their investment decisions [2]. A defining property of these strategic actors is their heterogeneous risk affinity, which is represented through individual discount rates (r_i). In this framework:

- A lower discount rate (e.g., $r = 0.08$) represents a more "patient" or risk-neutral investor willing to accept longer amortization periods to secure market shares, such as in the capacity-heavy aFRR market.
- A higher discount rate (e.g., $r = 0.15$ or 0.20) represents a more risk-averse investor who demands faster returns on investment, typically focusing on high-spread arbitrage opportunities in the spot market.

Finding a solution to an EPEC involves identifying a Nash Equilibrium, a state where no single investor can increase their profit by unilaterally changing their investment strategy. Given the non-linear and non-convex nature of coupled KKT conditions, the model is solved via an iterative diagonalization algorithm [2]. In this process, the MPEC of one investor is solved while the investment decisions of all other rivals are kept fixed. The algorithm cycles through all investors sequentially (Gauss-Seidel) or in parallel (Gauss-Jacobi) until the changes in investment decisions between iterations fall below a defined tolerance threshold, signifying that a stable equilibrium has been reached.

3.2 Upper Level: The Strategic Investor

At the Upper Level, each strategic investor $i \in I$ aims to maximize their individual NPV on a daily operational basis. The primary decision variables are the endogenously determined investment choices for power capacity ($X_{power,i,n}$) in MW and energy capacity ($X_{energy,i,n}$) in MWh at each node $n \in N$. The objective function maximizes the discounted profit from spot market arbitrage trading minus the specific per-day annualized capital expenditures (CAPEX) for power and energy capacity:

$$\max_{X_{power}, X_{energy}} NPV_i = \sum_{t,n} [\lambda_{n,t} \cdot (P_{discharge,i,n,t} - P_{charge,i,n,t})] - CAPEX_{daily,i} \quad (3.1)$$

To correctly internalize the CAPEX into the daily optimization loop, the investment costs are annualized using the Capital Recovery Factor (CRF). The CRF is calculated over the project lifetime (e.g., 15 years) utilizing the investor-specific discount rate (r_i):

$$CRF_i = \frac{r_i \cdot (1 + r_i)^{Lifetime}}{(1 + r_i)^{Lifetime} - 1} \quad (3.2)$$

The total investment cost is multiplied by the CRF and divided by the days of the year (365.25) to yield the exact daily CAPEX constraint:

$$CAPEX_{daily,i} = \frac{CRF_i}{365.25} \cdot \sum_n (C_{power} \cdot X_{power,i,n} + C_{energy} \cdot X_{energy,i,n}) \quad (3.3)$$

The investment decision is subject to several constraints:

1. **Shared Resource Constraint:** The total BESS power capacity installed at any given node n is restricted by a physical connection limit. Each investor must account for the fixed decisions of their rivals ($X_{power,j,n}^{fixed}$):

$$X_{power,i,n} + \sum_{j \neq i} X_{power,j,n}^{fixed} \leq Limit_n \quad (3.4)$$

2. **Energy-to-Power Ratio:** To ensure technical feasibility and align with typical battery configurations, a minimum ratio between energy and power capacity is enforced:

$$X_{energy,i,n} \geq Ratio_{min} \cdot X_{power,i,n} \quad (3.5)$$

3. **Equilibrium Constraints:** The investor's strategy is constrained by the response of the market clearing mechanism. This is mathematically integrated by including the KKT conditions of the Lower Level problem as constraints within the Upper Level.

3.3 Lower Level: Market Clearing

The Lower Level represents the ISO, who performs a market clearing to minimize the total dispatch costs of the conventional generators $g \in G$ over the time steps $t \in T$.

The objective function is defined as:

$$\min \sum_{t,g} MC_g \cdot P_{gen,g,t} \quad (3.6)$$

The market clearing process is subject to physical grid and technical storage constraints:

1. **Nodal Power Balance:** At each node n and hour t , the conventional generation, available wind/PV feed-in, and battery discharge must equal the nodal demand, net exports, and battery charge. The dual variable of this constraint represents the Locational Marginal Price ($\lambda_{n,t}$):

$$\sum_{g \in G_n} P_{gen,g,n,t} + P_{wind/pv,n,t} + \sum_i (P_{discharge,n,t} - P_{charge,n,t}) - D_{n,t} - NetExport_{n,t} = 0 \quad (3.7)$$

2. **Transmission Constraints:** To accurately represent physical power flows within the meshed network, the model employs a linearized DC Optimal Power Flow (DC-OPF) formulation. By utilizing a Power Transfer Distribution Factor matrix ($PTDF_{l,n}$), the active power flow on each transmission line l is computed directly from the net power injections at every node n . To ensure grid security, the resulting flow is strictly bounded by the predefined thermal transmission capacity limit ($Limit_l$):

$$\begin{aligned} NetInjection_{n,t} &= \sum_{g \in G_n} P_{gen,g,n,t} + P_{wind/pv,n,t} + \sum_i (P_{discharge,i,n,t} - P_{charge,i,n,t}) - D_{n,t} \\ Flow_{l,t} &= \sum_{n \in N} PTDF_{l,n} \cdot NetInjection_{n,t} \\ |Flow_{l,t}| &\leq Limit_l \end{aligned} \quad (3.8)$$

3. **Generator Limits:** Conventional generators are constrained by their maximum physical capacity:

$$0 \leq P_{gen,g,t} \leq P_{max,g} \quad (3.9)$$

4. **Storage Technical Equations:** The operation of the BESS is restricted by the capacities chosen at the Upper Level (X_{power}, X_{energy}) and governed by the State of

3 Methodology

Charge (SOC) balance equation, which accounts for the round-trip efficiency (η):

$$\begin{aligned}
0 &\leq P_{charge} \leq X_{power} \\
0 &\leq P_{discharge} \leq X_{power} \\
SOC_t &= SOC_{t-1} + \eta \cdot P_{charge,t} - \frac{1}{\eta} \cdot P_{discharge,t} \\
0 &\leq SOC_t \leq X_{energy} \\
SOC_{t=0} &= SOC_{t=24}
\end{aligned} \tag{3.10}$$

3.4 Model Expansions: Bridging the Research Gap

The systematic taxonomy developed in Section 2.1 identified critical gaps in existing BESS allocation models. While most models optimize BESS purely for day-ahead spot market arbitrage, real-world profitability heavily relies on multi-service market participation and is simultaneously constrained by physical asset degradation. To bridge this gap, the base MPEC was expanded to incorporate both battery degradation and the co-optimization of the spot and Automatic Frequency Restoration Reserve (aFRR) markets. To make the market more realistic, economic RES curtailment for negative prices was added. However, introducing these complex operational realities into a mathematical EPEC requires careful technical compromises to maintain solvability.

3.4.1 Battery Degradation

Battery life is significantly impacted by charge and discharge cycles. The literature review demonstrated that omitting degradation leads to overestimating the NPV, as intelligent operational strategies that minimize cyclic stress can substantially defer capital expenditure. Therefore, cycling costs must be internalized in the investor's objective function.

To incorporate this, the Upper Level objective function is adapted by subtracting the degradation costs ($Cost_{degrad}$) from the investor's operational revenues. Degradation is modeled as a linear marginal cost applied to the energy throughput. The specific mathematical element added to the objective function is defined as:

$$Cost_{degrad} = \sum_{t \in T} \sum_{n \in N} 0.5 \cdot C_{degrad} \cdot (P_{charge,i,n,t} + P_{discharge,i,n,t}) \tag{3.11}$$

In this new element, (C_{degrad}) represents the specific degradation cost parameter in €/MWh. The hourly charging (P_{charge}) and discharging ($P_{discharge}$) activities are summed

3.4 Model Expansions: Bridging the Research Gap

up and multiplied by a factor of 0.5, which accounts for the fact that a full battery cycle mathematically consists of both a charge and a discharge phase.

In reality, Lithium-Ion degradation is a highly non-linear process depending on factors like Depth of Discharge (DoD), SOC, and temperature, best captured by complex algorithms such as Rainflow-Counting as shown in [8]. However, Rainflow-Counting is non-differentiable. Because the EPEC implemented in this work relies on the Ipopt solver – which requires continuous, differentiable functions to solve the non-linear KKT systems – the Rainflow-Counting method had to be discarded. The linearized marginal cost approach serves as a necessary compromise to keep the optimization problem mathematically tractable while still penalizing excessive cycling.

3.4.2 Co-Optimization of Spot and aFRR Markets

BESS are technically predestined to provide frequency response due to their fast reaction times. Therefore optimal sizing for multiple electricity markets – such as participating in reserve markets alongside day-ahead markets – is a crucial next step for energy system modeling.

The model is expanded so that investors can withhold a portion of their capacity to provide positive (R_{up}) and negative (R_{down}) balancing power over standard 4-hour blocks ($b \in B$), based on the Austrian market design [17]. The Upper Level objective function is adapted to include the new revenues from the aFRR market ($Revenue_{afrr}$), leading to the final, comprehensive objective function for the strategic investor:

$$\max_{X_{power}, X_{energy}} NPV_i = Revenue_{spot} + Revenue_{afrr} - Cost_{degrad} - CAPEX_{daily} \quad (3.12)$$

The newly added aFRR revenue stream is calculated as:

$$Revenue_{afrr} = \sum_{b,n} H_{block} \cdot (\lambda_{afrr,up,b} \cdot R_{up,i,n,b} + \lambda_{afrr,down,b} \cdot R_{down,i,n,b}) \quad (3.13)$$

In this equation, H_{block} represents the duration of the respective block (4 hours in the Austrian case). The variables $R_{up,i,n,b}$ and $R_{down,i,n,b}$ are the strategic decisions on how much reserve capacity to provide. The terms $\lambda_{afrr,up,b}$ and $\lambda_{afrr,down,b}$ are the endogenously determined reserve clearing prices, which are derived from the Lower Level problem as dual variables. Initially, the aFRR price was determined by a KKT complementarity condition based on a deficit slack variable. However, early testing revealed that the non-linear solver exploited the mathematical relaxation boundaries, artificially driving the reserve price to its absolute price limit $Price_{cap}$ (e.g., 3,000 €/MW) even when providing near-zero capacity. To prevent this "price-dictator exploit", the aFRR clearing was reformulated into a continuous, deterministic Cournot

3 Methodology

Inverse Demand Curve. The endogenous clearing price (λ_{afrr}) is directly dependent on the aggregated capacity offered by all market participants relative to the exogenous demand:

$$\begin{aligned}\lambda_{afrr,up,b} &= Price_{cap} \cdot \left(1 - \frac{\sum_n R_{up,i,n,b} + \sum_{j \neq i,n} R_{up,j,n,b}^{fixed}}{Demand_{up,b}} \right) \\ \lambda_{afrr,down,b} &= Price_{cap} \cdot \left(1 - \frac{\sum_n R_{down,i,n,b} + \sum_{j \neq i,n} R_{down,j,n,b}^{fixed}}{Demand_{down,b}} \right)\end{aligned}\quad (3.14)$$

If investors fail to provide capacity, the price peaks at the $Price_{cap}$ of 3,000 €/MW. If they fulfill the demand exactly, the market is saturated and the price drops to zero. This forces the investors into a realistic, competitive oligopoly. This co-optimization introduces an opportunity cost logic: providing reserve capacity restricts the available bandwidth for spot market arbitrage. Consequently, the technical storage constraints are tightened to reflect that the sum of spot power and reserved capacity must not exceed the installed power capacity (X_{power}):

$$\begin{aligned}P_{charge,t} + R_{down,b} &\leq X_{power} \\ P_{discharge,t} + R_{up,b} &\leq X_{power}\end{aligned}\quad (3.15)$$

3.4.3 Economic RES curtailment

To prevent negative spot prices during periods of extreme renewable energy surplus, an economic curtailment feature was introduced. Modeling curtailment via traditional KKT complementarity conditions (e.g., $RES_{curtail} \cdot \lambda_{spot} \leq \epsilon$) within a decentralized MPEC framework led to severe solver exploits. The optimizer utilized the tiny mathematical tolerance (ϵ) to feign microscopic curtailment, artificially forcing the local MPEC spot price to 0 €/MWh, allowing the BESS to charge for free while shifting the actual systemic costs to peak generators.

To resolve this zero-price exploit, the bi-linear KKT approach for curtailment was discarded. Instead, curtailment is managed globally via strict dual optimality: The nodal spot price is strictly bounded above zero ($\lambda_{n,t} \geq 0$), and a microscopic penalty (e.g., 0.01 €/MW) for curtailment is subtracted directly from the investors' objective function. Consequently, the market organically curtails excess generation only when the spot price naturally hits the 0 €/MWh floor, entirely removing the solver's ability to manipulate non-linear KKT constraints.

3.4.4 Discarded model features

To formulate a stable and solvable EPEC, several other real-world market characteristics had to be consciously excluded:

1. **Integer Bidding (Minimum Bid Sizes):** Real aFRR markets often require bids in discrete steps (e.g., exactly ± 1 MW) [17]. This constraint was discarded, and capacity investments as well as aFRR dispatch are modeled continuously. Formulating this as a Mixed-Integer Program with Equilibrium Constraints (MIPEC) using binary variables would have caused numerical instability, rendering the problem nearly impossible to solve with the utilized Interior Point optimizer (Ipopt). This continuous relaxation is justified by assuming that the strategic investor acts as an aggregator, utilizing a pool of batteries to smooth out granular bidding limits.
2. **Stochasticity and Balancing Energy:** The model assumes "perfect foresight", meaning future prices, wind, and solar feed-in are known deterministically. Introducing stochastic optimization to account for forecast errors and subsequent balancing energy activation was discarded. The computational burden of evaluating multiple stochastic scenarios within an iterative EPEC diagonalization loop would have led to an explosion in computation time, making the framework unsolvable within a reasonable timeframe.

3.5 EPEC Solution Strategy and Algorithm

Solving an EPEC mathematically translates to finding a Nash Equilibrium. In this state, no single strategic investor can unilaterally increase their profit by changing their investment strategy, given that the strategies of all other competitors remain constant [2]. Because the individual MPECs of the investors are intrinsically coupled through the market clearing prices (LMPs) and the shared grid constraints, they cannot be solved isolated in a single step. Instead, an iterative approach is required.

3.5.1 The Diagonalization Algorithm

To resolve the interdependencies between the strategic actors, a Diagonalization Algorithm is employed. This iterative approach is well-established and widely accepted in the state-of-the-art literature for solving and verifying EPEC in energy market modeling [18, 16]. The core logic of this algorithm relies on an iterative loop structure: Within a defined maximum number of overall iterations, the algorithm iterates through the set of all investors. When solving the MPEC for investor i to determine their optimal

3 Methodology

capacity allocation, the investment decisions of all other rival investors $j \neq i$ are treated as fixed, exogenous parameters based on the results of previous calculations. Once investor i 's MPEC is solved and their new capacities are determined, the algorithm proceeds to the next investor, repeating this process until a stable state is reached.

3.5.2 Convergence Criteria and Multiple Equilibria

A critical aspect of EPEC modeling is defining when an equilibrium is reached. In this framework, convergence is evaluated by continuously tracking the maximum change in the main decision variable – the installed power capacity (X_{power}) – across all investors and nodes between two consecutive iterations. This metric is denoted as *Max Delta*. The algorithm is set to terminate when the Max Delta falls below a predefined tolerance threshold (e.g., 0.5 MW). If the capacity adjustments of all investors fall below this margin, the market has reached a stable Nash Equilibrium.

However, due to the non-linear, non-convex nature of EPECs and the presence of shared network constraints, the mathematical solution space is highly complex. Consequently, EPECs in power markets often do not possess a single unique solution, but rather exhibit multiple pure-strategy equilibria. The existence of multiple equilibria, resulting in varied investment decisions and profit distributions depending on algorithmic starting points, is a well-documented phenomenon in EPEC literature . Depending on the initial starting values and algorithmic tuning parameters, the model may converge to different, yet equally valid, stable market states [18, 16].

3.5.3 Sequential vs. Parallel Execution

The order in which the investors' MPECs are solved within the diagonalization loop significantly impacts the solution process. The framework allows for two distinct mathematical execution strategies:

1. **Gauss-Seidel (Sequential):** In this traditional approach, the MPECs are solved sequentially. As soon as investor i updates their strategy, this new strategy is immediately fixed and visible to the next investor $i + 1$ within the same iteration step. While mathematically robust, this sequential nature creates a computational bottleneck, as each NLP must be solved one after the other.
2. **Gauss-Jacobi (Parallel):** To overcome computational limitations, the algorithm was transitioned to a Gauss-Jacobi scheme. In this approach, all investors calculate their optimal strategy for iteration k simultaneously, based purely on the fixed, global market state from the previous iteration $k - 1$.

3.5 EPEC Solution Strategy and Algorithm

From a technical perspective, the Gauss-Jacobi method allows for the parallelization of the optimization problem. Implemented in Python, the *ProcessPoolExecutor* module is utilized to spawn separate, simultaneous processes for each investor's MPEC. This ensures that the heavy computational load is distributed across multiple CPU cores, drastically reducing the real-time duration of an iteration cycle.

3.5.4 Technical Tweaks for Computational Speed

Solving an EPEC is mathematically arduous because it requires solving a series of NLP problems filled with complex KKT complementarity constraints. As highlighted in recent literature, the sheer computational complexity of EPECs makes them notoriously difficult to solve, often requiring algorithmic workarounds or high-performance solvers to maintain tractability [18].

Initial implementations revealed that the default solvers struggled with the sheer size of the matrices, resulting in low CPU utilization during the NLP solution process. To significantly accelerate the process, the algorithm was technically optimized by integrating the *CoinHSL mathematical software library* [19], specifically utilizing the high-performance linear solver *MA97* within the Ipopt optimization framework. Combined with forcing multi-threading logic (*OMP_NUM_THREADS = 16*) and the parallelized Gauss-Jacobi execution, the computational time to solve four highly complex, co-optimized MPECs (including aFRR markets) was reduced to approximately 5 to 15 seconds per iteration. This technical tweaking allows the entire EPEC to reliably find a Nash Equilibrium within roughly 3 minutes.

3.5.5 Model Tweaks for Stability and Tractability

Modeling capacity expansion equilibria in liberalized electricity markets using an EPEC approach is notoriously challenging due to the inherent mathematical complexity of strategic multi-leader-follower games. The inclusion of the Lower Level's KKT conditions into the Upper Level introduces strong non-linearities and non-convexities, often making it difficult to find stable market equilibria without algorithmic adjustments [16, 3]. To ensure that the Python-based optimization framework using the Ipopt solver terminates successfully and yields economically meaningful results, several empirical model tweaks and stabilization mechanisms were implemented.

3 Methodology

Damping Factor against Algorithmic Oscillation

Iterative diagonalization algorithms are prone to severe oscillations. During the initial EPEC loops, it was observed that strategic investors would alternately over-invest and under-invest at highly profitable nodes in consecutive iterations, creating a "zig-zag" behavior that prevented the Max Delta convergence criterion from being met. To stabilize the solution path, a mathematical Damping Factor (α) was introduced into the capacity update rule. Instead of immediately adopting the newly calculated optimal capacity ($X_{calculated}$) for the next iteration, the actual capacity applied (X_{new}) is a weighted average of the previous iteration's state (X_{old}) and the new result:

$$X_{new} = (1 - \alpha) \cdot X_{old} + \alpha \cdot X_{calculated} \quad (3.16)$$

By setting α strictly below 1 (e.g., $\alpha = 0.25$), the aggressiveness of the strategic responses is artificially slowed down, guiding the system smoothly towards a Nash Equilibrium.

Soft Constraints and Deficit Mechanisms

A common cause for solver failure in EPECs is the strictness of shared physical constraints, which can lead to mathematically infeasible regions. For example, if multiple strategic investors simultaneously attempt to provide large quantities of balancing power, the rigid lower-level market constraints can be violated. In real-world market models, extreme scarcity leads to price spikes, whereas in raw mathematical models, it simply leads to an abrupt termination or dual variables exploding to infinite bounds (e.g., 10^{16}). To remedy this, strict limits were relaxed using Soft Constraints and slack variables. Most prominently, a "Deficit" variable was introduced for the aFRR market demand. Instead of forcing the market strictly to meet the reserve demand, the model is allowed to fall short by activating the deficit variable at a predefined, exceptionally high penalty cost (Price Cap). This ensures that the feasible region of the optimization problem remains intact, bounding the endogenous market prices logically while still penalizing under-provision.

Relaxation of KKT Complementarity

The strict complementarity conditions of the Lower Level market clearing (where the product of two variables must equal exactly zero) are entirely non-smooth. While some mixed-integer linear programming (MILP) approaches resolve this using "Big-M" constraints with binary variables, this EPEC utilizes a non-linear interior-point solver (Ipopt). Thus, strict complementarities were slightly relaxed using a tolerance value

3.6 Input Parameters and Empirical Scale

ϵ (e.g. 10, determined empirically). This continuous relaxation smooths the feasible region, preventing the solver from getting trapped in local, infeasible mathematical corners during early iterations, while still strictly enforcing economic rationality as the algorithm converges. While this relaxation introduces minor numerical noise into the results (e.g., allowing a generator to output 0.25 MW despite a 0 €/MWh price due to the 10 slack allowance), this noise is economically insignificant.

BESS Residual Demand

Initial implementations of the diagonalization algorithm fixed the specific charging and discharging variables of competitor j and passed them into the Lower Level problem of investor i . However, this caused severe "cobweb" oscillations and negative arbitrage, as investors blindly evaded each other's schedules without correctly anticipating the spot price collapse.

To overcome this, a Residual Demand formulation was implemented. Instead of representing all competitors within the active investor's MPEC, the aggregated actions of all rival investors ($j \neq i$) from the previous iteration are pre-processed into a modified systemic residual load ($D_{adj,n,t}$):

$$D_{adj,n,t} = D_{n,t} + \sum_{j \neq i} (P_{charge,j,n,t}^{fixed} - P_{discharge,j,n,t}^{fixed}) \quad (3.17)$$

By optimizing their own dispatch strictly against this adjusted residual demand, the active investor accurately anticipates the market price shifts caused by their competitors. This fundamental game-theoretic restructuring stabilizes the Nash Equilibrium and ensures rational, profitable arbitrage behavior.

3.6 Input Parameters and Empirical Scale

To generate realistic scarcity signals and robust economic incentives for the strategic BESS investors, the mathematical model must be parameterized with empirical data. The configuration of the grid, generation fleet, and storage characteristics follows established state-of-the-art literature while being tailored to a representative single-day horizon.

3 Methodology

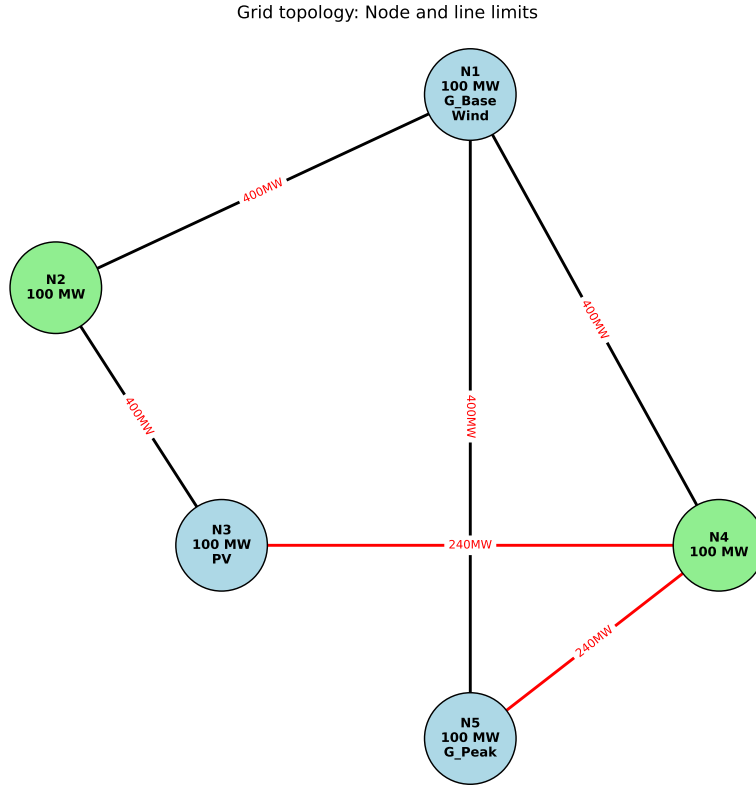


Figure 3.2: 5-bus test system with node and line limits. G.Base refers to the conventional generators for base load, G.Peak refers to the conventional generators for peak load.

3.6.1 Grid Topology and Generation Fleet

While many studies rely on large-scale transmission networks such as the IEEE 33-bus or 39-bus systems [7, 13], solving an EPEC is computationally highly restrictive. To balance computational tractability with the necessity of representing a meshed network with loop flows, the modified 5-bus test system seen in Figure 3.2 is employed.

The transmission limits of the connecting lines are strictly bounded (e.g., between 240 MW and 400 MW) to provoke network congestion, which serves as a primary driver for locational price spreads. The conventional generation fleet is divided into base-load and peak-load units, strategically placed across the nodes. To establish a distinct and realistic merit-order curve, these units are assigned specific marginal generation costs (variable costs in €/MWh). In the techno-economic parameters provided by Devine and Siddiqui [18], existing base-load units operate at approximately 48.87 €/MWh, whereas

3.6 Input Parameters and Empirical Scale

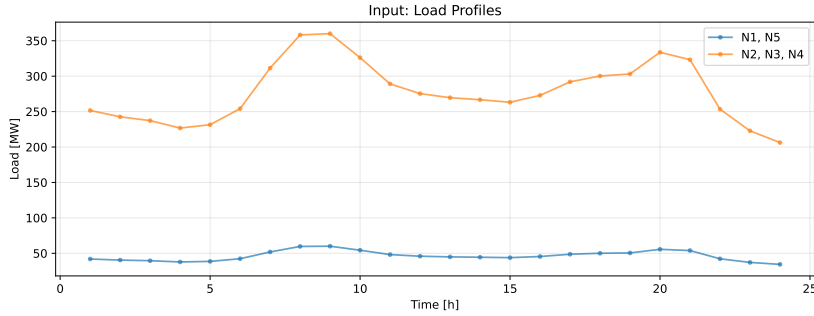


Figure 3.3: Load Profiles per Node.

the more expensive peak-load units are assigned marginal costs of 63.38 €/MWh. For this calibrated model, the marginal generation costs are set to 40 €/MWh for base-load and 80 €/MWh for peak-load to ensure sufficient price spreads for realistic battery arbitrage.

3.6.2 Demand and Renewable Energy Profiles

The temporal parameters are based on real-world 24-hour profiles of the representative day 11.09.2025, extracted from Austrian TSO Austrian Power Grid (APG) data [20]. To observe realistic battery cycling behavior, the profiles include:

- **Load Profile:** Scaled to a system-wide peak load of 1200 MW, exhibiting a typical “duck curve” with distinct morning and evening demand peaks, as seen in Figure 3.3. The load is spatially distributed, with 90% concentrated in the central nodes (N2, N3, N4) and the remaining 10% in the peripheral nodes (N1, N5).
- **Renewable Generation:** The model incorporates a highly volatile wind power profile with a peak capacity of 370 MW concentrated at node N1 (generating primarily at night and early morning), and a typical photovoltaic (PV) bell curve with a peak of 430 MW at node N3.

As demonstrated by Czock et al. [15], the spatial discrepancy between renewable generation centers and high-demand nodes is a crucial factor for efficient storage allocation, which this empirical setup explicitly mimics.

3.6.3 BESS Characteristics and Costs

A critical aspect of the BESS allocation problem is the correct economic assessment of the storage technology. The current literature underscores the high initial capital

3 Methodology

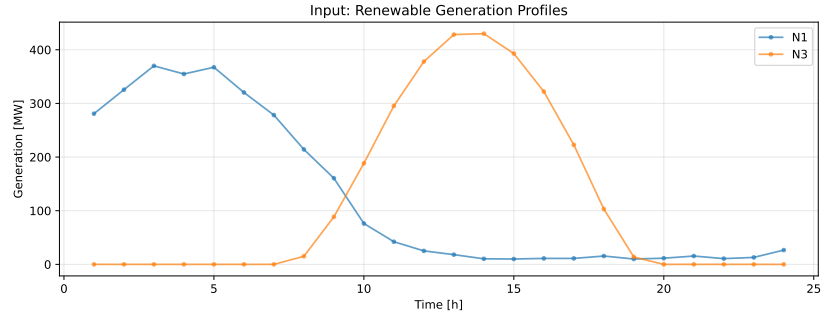


Figure 3.4: Renewable Generation Profiles per Node. N1 employs Wind whereas N3 employs PV.

expenditure (CAPEX) of Lithium-Ion (Li-Ion) BESS, making accurate sizing essential for project profitability.

In this framework, the investment costs are split into power capacity costs (C_{power} , representing the Power Conversion System) and energy capacity costs (C_{energy} , representing the battery cells). To internalize these capital costs into the single-day optimization loop, the total CAPEX is annualized over the expected project lifetime (e.g., utilizing a 15-year horizon as common in BESS projects). Based on current techno-economic projections for utility-scale Li-Ion systems, this results in annualized daily cost parameters set at 6,600 €/MW for power and 18,800 €/MWh for energy capacity. [8].

To ensure technical feasibility and reflect realistic utility-scale storage configurations, a minimum E/P ratio constraint ($Ratio_{min}$) is strictly enforced at the Upper Level, as described in Equation 2. In this framework, the minimum E/P ratio is set to 2 hours. This establishes a balanced and widely adopted industry standard for systems performing multi-service applications [7, 8].

To accurately capture the physical limitations of the storage assets, several technical constants are integrated into the lower-level problem. Based on the specifications by Shin and Hur, the overall round-trip efficiency (η) of the BESS is modeled at 93.6% [8].

As battery capacity degradation significantly impacts long-term profitability, the degradation penalty described in Section 3.4.1 is enforced. Assuming typical Li-Ion characteristics (e.g., a cycle life of 3,500 to 6,000 cycles), this is parameterized at 15 €/MWh throughput, accurately forcing the BESS to internalize its physical wear during operations.

4 Results & Discussion

This chapter presents the quantitative results of the developed EPEC modeling framework. To comprehensively evaluate the impact of multi-service market participation on the strategic allocation of BESS, a comparative analysis is conducted between two distinct market setups: an **Energy-Only Scenario** (focused solely on day-ahead spot market arbitrage) and a **Multi-Service Scenario** (co-optimizing spot arbitrage with continuous aFRR provision).

The Energy-Only setup serves as a benchmark, reflecting the focus of conventional BESS allocation models. By explicitly contrasting it with the Multi-Service setup, this comparison aims to isolate and quantify the specific effects of reserve market obligations on strategic investment decisions. The objective of this comparison is to demonstrate how lucrative capacity premiums and the resulting power bottlenecks fundamentally alter optimal BESS sizing.

The remainder of this chapter is organized as follows. Following a brief evaluation of the algorithmic convergence and stability of the EPEC framework, Section 4.2 analyzes the strategic investment behavior regarding spatial distribution (siting), power capacity (MW), and the resulting Energy-to-Power (E/P) ratios. Section 4.3 breaks down the distinct revenue streams and investigates the spillover effect of the aFRR market on the arbitrage market. Finally, the chapter evaluates the aggregated system balance and nodal price dynamics, illustrating how multi-service obligations synchronize BESS operations across structurally diverse network nodes.

4.1 Algorithmic Performance and EPEC Stability

As detailed in Section 3.5, solving the EPEC relies on a parallelized Gauss-Jacobi diagonalization algorithm, with convergence achieved when the *Max Delta* of the installed capacity falls below 0.5 MW. The practical execution of both scenarios demonstrates that the model tweaks introduced in Section 3.5.5 were strictly necessary to stabilize the framework. Figure 4.1 illustrates the convergence history for both the Energy-Only and the Multi-Service scenarios, tracking the *Max Delta* metric of the installed capacity.

4 Results & Discussion

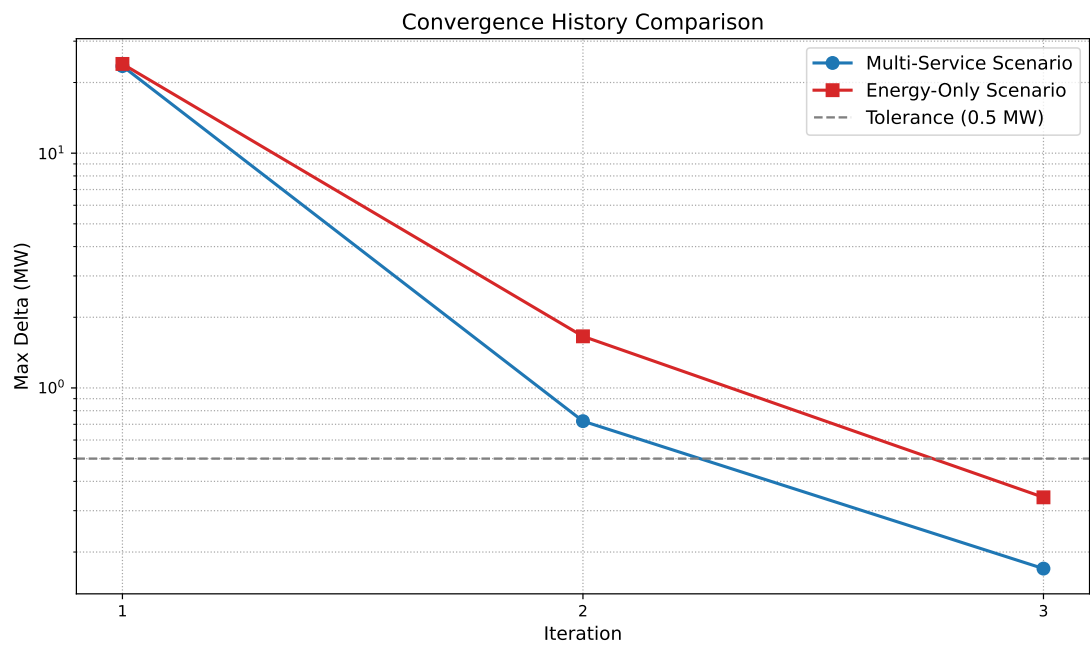


Figure 4.1: Convergence history comparison displaying the *Max Delta* in MW per iteration.

4.2 Strategic BESS Allocation: Siting, Sizing, and the E/P Ratio

In contrast to typical EPEC behaviors documented in the literature, where pure arbitrage trading often leads to continuous "zig-zag" oscillations, the Energy-Only Scenario converged exceptionally fast. By optimizing strictly against the pre-calculated residual load of their rivals, the strategic investors accurately anticipated market price shifts. Combined with a damping factor of 0.25, the algorithm reliably reached a stable Nash Equilibrium in only 3 iterations, well below the strict tolerance threshold of 0.5 MW. The Multi-Service Scenario exhibited identical, rapid convergence. Despite the additional mathematical stiffness introduced by the simultaneous co-optimization of strict state-of-charge constraints for aFRR blocks, the model terminated optimally after 3 iterations. The replacement of rigid KKT-deficit rules with the continuous Cournot pricing mechanism eliminated all previously observed exploits, seamlessly guiding the competitive oligopoly into a stable and mathematically sound equilibrium.

4.2 Strategic BESS Allocation: Siting, Sizing, and the E/P Ratio

The transition from a pure energy arbitrage market to a multi-service environment fundamentally alters the strategic investment behavior of the agents. By examining the results of the Energy-Only and Multi-Service scenarios, distinct patterns in spatial distribution (siting), power capacity (sizing in MW), and energy capacity (sizing in MWh) emerge.

In both scenarios, the total installed power capacity approaches the theoretical physical limits of the network. The shared connection limit of 100 MW per node is fully utilized across all five nodes, resulting in a total aggregated capacity that closely approaches the system-wide maximum of 500 MW. The perfectly balanced spatial distribution and the absence of any nodal overbooking demonstrate that the solver correctly and strictly enforces the hard boundaries of the shared resource constraints.

Furthermore, the continuous Cournot Inverse Demand curve establishes a highly competitive and balanced oligopoly. As detailed in Table 4.1, the power capacity provision is shared almost equally among all participants in the Multi-Service scenario, with each investor installing approximately 124 MW. The varying discount rates (cost of capital ranging from 8% to 20%) do not lead to the exclusion of any participant from the power capacity market. Instead, the investors strategically share the capacity provision to maintain high endogenous clearing prices, deliberately avoiding the price collapse that would occur if a single investor oversupplied the reserve market.

While the nominal power capacity (MW) is evenly distributed among the competitors, the fundamental impact of the investors' heterogeneous risk affinity (represented

4 Results & Discussion

by their individual discount rates) becomes highly evident in their energy capacity (MWh) sizing. As established in the methodology (Section 3.6.3), a minimum Energy-to-Power (E/P) ratio of 2.0 hours is enforced. However, the endogenously optimized configurations deviate significantly from this baseline, especially when multi-service requirements are introduced.

In the Multi-Service Scenario, the simultaneous co-optimization of 4-hour aFRR blocks and spot market arbitrage requires substantial energy reservoirs to compensate for the severe power bottleneck created by continuous reserve provision. Because the capacity is blocked for the highly profitable aFRR market, the batteries are stripped of their operational flexibility. Investor 1, possessing the lowest cost of capital ($r = 0.08$), leverages its financial advantage to install a massive energy capacity of 2,648.58 MWh, resulting in an exceptionally high E/P ratio of 21.21 hours. Overlooking the unrealistic project size, this “patient” investor can amortize the immense upfront CAPEX for cheap battery cells over the 15-year project lifetime to extract marginal revenues out of the spot market with its remaining fractional power capacity.

Conversely, investors with higher capital costs cannot afford such capital-intensive energy reservoirs. Investor 2 ($r = 0.12$), for instance, restricts its energy capacity to 291.13 MWh, resulting in an E/P ratio of 2.35 hours—staying very close to the technical minimum to minimize investment costs. This empirical finding highlights a crucial dynamic in liberalized multi-service storage markets: the cost of capital is the primary driver for the depth of storage (energy capacity) rather than the nominal power capacity (MW) in an oligopolistic setting.

Table 4.1: Aggregated BESS investment metrics comparing the Energy-Only and Multi-Service scenarios.

Investor	WACC	Energy-Only Scenario			Multi-Service Scenario		
		Total MW	Total MWh	E/P Ratio	Total MW	Total MWh	E/P Ratio
Investor 1	8.0%	122.22	484.32	3.96	124.90	2648.58	21.21
Investor 2	12.0%	126.57	2392.83	18.91	123.62	291.13	2.35
Investor 3	15.0%	124.42	758.93	6.10	123.99	716.09	5.78
Investor 4	20.0%	123.90	595.77	4.81	124.57	625.89	5.02

4.3 Revenue Stream Breakdown and the Arbitrage-Reserve Trade-off

The allocation of generation capacity is fundamentally driven by the expected profitability of the assets. Analyzing the revenue streams—specifically the distribution between

4.3 Revenue Stream Breakdown and the Arbitrage-Reserve Trade-off

day-ahead spot market arbitrage and aFRR capacity provision—reveals critical insights into the operational priorities forced upon the BESS by the market design.

In the Energy-Only Scenario, the model demonstrates economically sound and rational arbitrage behavior. Given the calibrated system parameters (marginal generation costs of 40 €/MWh for base-load and 80 €/MWh for peak-load, alongside a degradation cost of 15 €/MWh), the price spreads are sufficient to offset efficiency losses and physical battery wear. As derived from the simulation logs, all four investors achieve solid, positive daily arbitrage profits ranging from approximately 52,600 € (Investors 3 and 4) to 68,200 € (Investor 2). The BESS assets act as pure price-takers in the time-shifting of energy, buying during low-price periods (high wind/PV feed-in) and discharging during peak residual load.

However, the introduction of the reserve market in the Multi-Service Scenario triggers a massive paradigm shift in revenue generation. Driven by the continuous Cournot Inverse Demand curve, the strategic investors form a highly profitable oligopoly in the aFRR market. By collectively withholding capacity to intentionally fulfill only a portion of the system operator's total reserve demand, the strategic actors keep the endogenous aFRR clearing price artificially high (near 600 €/MW per hour). Consequently, the primary source of income entirely shifts to the reserve market. As detailed in Table 4.2, daily aFRR revenues reach between 292,986 € and 293,041 € for Investors 2, 3, and 4. Investor 1, leveraging its massive energy capacity (E/P ratio of 21.21) and low capital costs, captures an even higher aFRR revenue of 332,852 €.

The overwhelming financial dominance of the reserve market comes at a severe operational cost, manifesting as stark arbitrage-reserve trade-off. While Investors 2, 3, and 4 see their spot market profits plummet to a fraction of their previous values (between 3,500 € and 4,500 €), Investor 1 deliberately operates its BESS at a spot market loss, generating a arbitrage loss of -2,574 € per day.

This counter-intuitive behavior is driven by a cross-market opportunity cost logic caused by a severe power bottleneck. Due to the high margins dictated by the Cournot inverse demand, investors strategically allocate almost their entire expensive inverter capacity (MW) to reserve provision. To still capture any remaining arbitrage spreads, investors drastically oversize their relatively cheap battery cell capacity (MWh), resulting in extraordinarily high Energy-to-Power (E/P) ratios (e.g., > 21 hours). The loss-making spot market operation is a direct consequence of this inverter capacity lock-up: the batteries are forced into highly inefficient, prolonged charge and discharge cycles just to squeeze marginal revenues out of the remaining fractional MW capacity, accepting operational spot market losses to sustain the overall profit-maximizing strategy.

Ultimately, the strategic investors willingly sacrifice spot market profitability, accepting negative arbitrage as an unavoidable opportunity cost necessary to unlock the vastly

4 Results & Discussion

superior cartel profits of the aFRR oligopoly.

Table 4.2: Daily revenue stream breakdown comparing Arbitrage and aFRR profits across the Energy-Only and Multi-Service scenarios.

Investor	WACC	Energy-Only Scenario		Multi-Service Scenario	
		Arbitrage (€)	aFRR (€)	Arbitrage (€)	aFRR (€)
Investor 1	8.0%	64,660.66	0.00	-2,574.26	332,852.44
Investor 2	12.0%	68,297.04	0.00	4,521.00	292,986.68
Investor 3	15.0%	52,641.06	0.00	3,759.11	293,034.14
Investor 4	20.0%	52,686.17	0.00	3,532.13	293,041.04

4.4 System-Wide Operation Profiles and Grid Congestion

Beyond financial revenue maximization, the strategic operation of the BESS portfolio has a profound physical impact on the power system. Analyzing the aggregated system balances (Figure 4.2) reveals how the BESS units dynamically adapt to renewable energy intermittency and diurnal demand patterns.

The temporal generation profiles in the test system create four distinct operational phases over the 24-hour period. During the night and early morning (hours 1 to 6), the system experiences high wind power feed-in coupled with low system demand. As observed in the system balance, the BESS portfolio aggressively charges during these hours (indicated by the negative net BESS power). This coordinated charging absorbs the excess renewable generation that would otherwise force the base-load generator to ramp down unnecessarily.

As the day progresses into the morning peak (hours 7 to 9), the system encounters its absolute daily maximum in load demand. Correspondingly, the BESS portfolio shifts into its primary discharging phase, reaching its maximum injection power for the day. This critical intervention supplies the high morning demand just before solar generation fully ramps up, actively preventing severe capacity shortages.

During the mid-day period (hours 10 to 15), the system is characterized by strong photovoltaic (PV) generation, significantly suppressing the residual load. The BESS units utilize this secondary low-price window to aggressively recharge and restore their state-of-charge.

Conversely, during the evening peak (hours 17 to 21), when demand experiences a secondary surge and solar generation ceases, the BESS portfolio enters its second

4.5 Nodal Price Dynamics and Local Arbitrage

discharging phase. While the injection power is lower than during the morning peak, this operation remains vital. By injecting previously stored renewable energy into the grid, the storage units displace expensive peak-load generation, effectively capping the system-wide price spikes and supplying the evening demand.

While the fundamental time-shifting behavior is present in both scenarios, the Multi-Service Scenario introduces observable rigidities. In the Energy-Only setup, BESS operations perfectly mirror the spot price spreads, resulting in deep, continuous charge and discharge cycles. However, the requirement to provide continuous 4-hour reserve capacity blocks (aFRR) forces the BESS into shallower and more constrained spot market operations, as the batteries must maintain sufficient energy reservoirs for potential reserve activation.

These system-wide operations are intrinsically linked to the physical limitations of the transmission network. The accompanying line loading heatmaps (Figure 4.3) illustrate the spatial grid utilization. Notably, transmission lines L14, L15, and L34 operate well below their thermal limits throughout the day. Instead of forcing power through constrained corridors, the previously discussed optimal sizing strategy (see Section 4.2)—where BESS capacity is evenly distributed across all five nodes—acts as a critical congestion relief mechanism. By charging locally during periods of excess generation and discharging locally during peak demand, the BESS portfolio effectively balances the system without pushing these transmission lines to their limits.

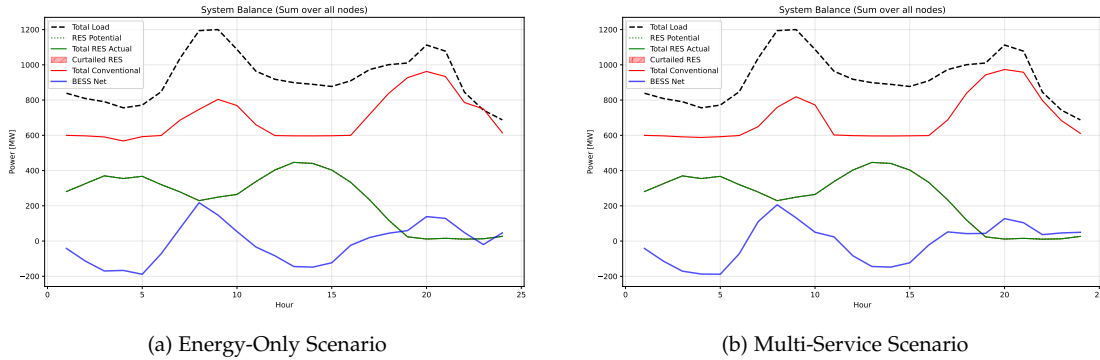


Figure 4.2: Aggregated system balance over 24 hours. The solid blue line represents the net BESS dispatch (negative values indicate charging mode).

4.5 Nodal Price Dynamics and Local Arbitrage

The aggregated system behavior discussed in the previous sections is fundamentally driven by the underlying Locational Marginal Prices (LMPs). In the decentralized MPEC

4 Results & Discussion

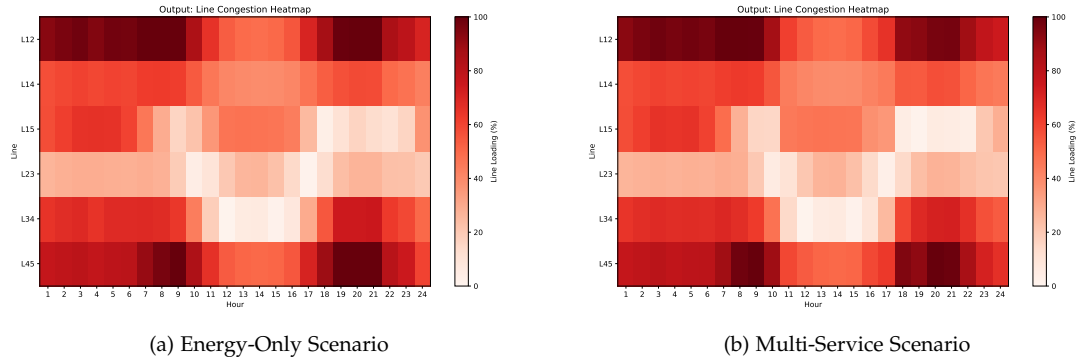


Figure 4.3: Heatmaps of transmission line loading. Lines L14, L15, L23 and L34 remain structurally unconstrained, demonstrating the effectiveness of locally distributed BESS capacities.

framework, these nodal prices act as the primary scarcity signals, coordinating the competitive local arbitrage strategies of the BESS investors. Analyzing the individual nodal balances (Figures 4.4 and 4.5) reveals extreme spatial price spreads generated by the heterogeneous distribution of generation assets and strict transmission limits.

In the Energy-Only Scenario, the price dynamics exhibit extreme local divergence. At Node 3 (PV-dominated), the LMP demonstrates severe scarcity pricing during the morning and evening system peaks, spiking from the 80 €/MWh peak-load marginal cost to over 600 €/MWh. The BESS portfolio logically reacts to these massive arbitrage opportunities by discharging heavily during these exact hours.

Conversely, Node 1 (Wind and Base-Load) demonstrates a paradoxical price dynamic during the very same periods. While the rest of the system experiences peak demand and high prices, the local LMP at Node 1 drops from 80 €/MWh down to 40 €/MWh during the morning and evening peaks. This phenomenon is a textbook result of nodal pricing under structural grid congestion: the transmission line L12 hit their thermal limits, effectively isolating the node and trapping the cheap 40 €/MWh base-load generation locally. Despite these unexpectedly low local prices during peak hours, the BESS units still perform minor discharging operations, indicating complex interactions with their internal state-of-charge cycles.

While the Energy-Only setup highlights isolated local market conditions, the Multi-Service Scenario fundamentally disrupts this spatial divergence. The commitment to 4-hour aFRR blocks forces a systemic synchronization of BESS operations, which consequently homogenizes the nodal prices.

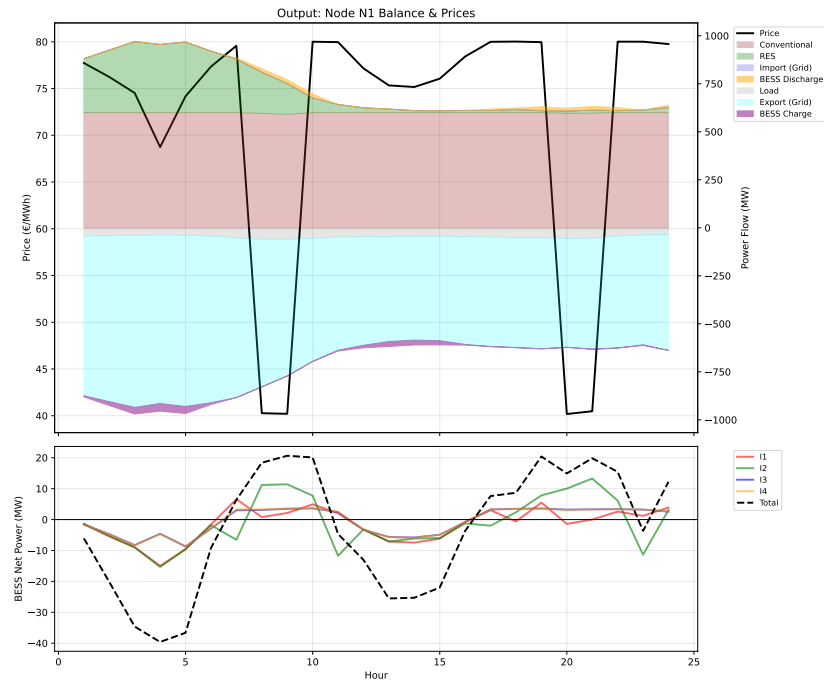
This synchronization is clearly visible when comparing the operational strategies at Node 3 and Node 1. At Node 3, the morning price settles near 50 €/MWh, prompting the BESS to charge. The price then climbs to 80 €/MWh and holds until noon, triggering

4.5 Nodal Price Dynamics and Local Arbitrage

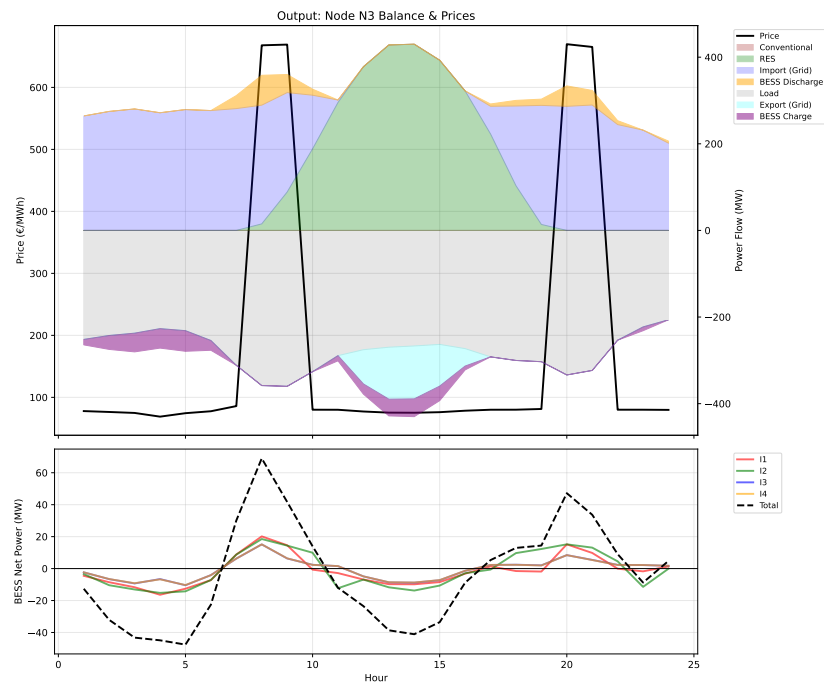
BESS discharge. As the mid-day PV peak hits, the price sharply drops to 40 €/MWh, creating a brief charging window, before surging back to 80 €/MWh at 16:00 due to high evening demand, leading to the final discharge phase.

Remarkably, the price fluctuations and the BESS charging strategy at Node 1 mirror those at Node 3 almost identically. This demonstrates that the rigid state-of-charge requirements of the system-wide aFRR market completely dominate the operational logic of the batteries. The localized network constraints and nodal scarcity signals, which defined the Energy-Only scenario, are largely overridden by the lucrative, yet inflexible, obligations of the reserve market.

4 Results & Discussion



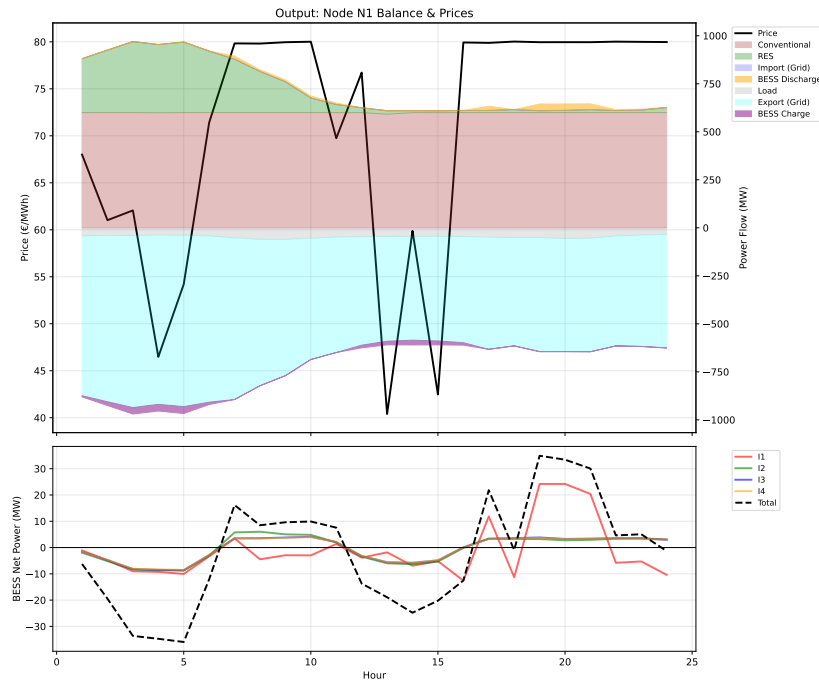
(a) Node 1 (Energy-Only)



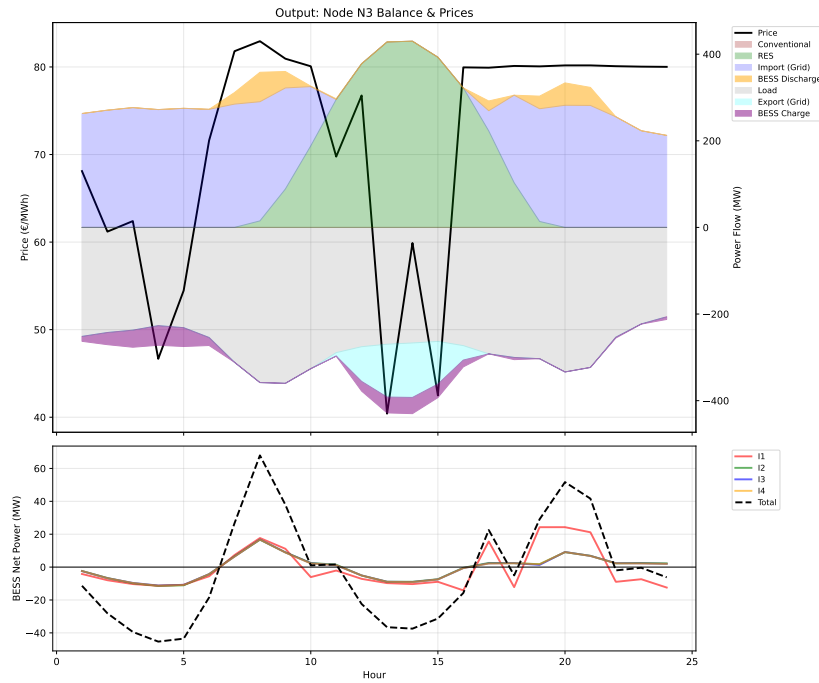
(b) Node 3 (Energy-Only)

Figure 4.4: Nodal balances in the Energy-Only Scenario. Notice the scarcity price spikes at N3 and the paradoxical price drops at N1 during system peaks.

4.5 Nodal Price Dynamics and Local Arbitrage



(a) Node 1 (Multi-Service)



(b) Node 3 (Multi-Service)

Figure 4.5: Nodal balances in the Multi-Service Scenario. The aFRR obligations synchronize BESS operations and price dynamics across structurally different nodes.

5 Synthesis of the Results

While Chapter 4 provided a detailed quantitative analysis of the simulation data, this chapter elevates the findings to a broader meta-level. The purpose of this synthesis is to directly answer the core research question, critically evaluate the strengths and limitations of the applied EPEC methodology, discuss the transferability of the findings to real-world power systems, and reflect on the lessons learned during the modeling process.

5.1 Answers to the Research Question

The overarching research question of this thesis explored what insights into strategic investor behavior can be gained by shifting from a traditional central-planner cost-minimization approach to a decentralized, game-theoretic EPEC framework. Based on the quantitative results, this question can be answered clearly: Traditional cost-minimization models implicitly assume perfect system-wide coordination and unselfish asset deployment. In contrast, the EPEC framework uncovered that strategic, profit-maximizing behavior fundamentally alters both the operational logic and the physical dimensioning of BESS.

Regarding the **siting (location)** of BESS, the results demonstrate that strategic investors distribute their assets across the entire grid. Rather than clustering exclusively at nodes with the highest renewable generation (e.g., wind at Node 1 or PV at Node 3), the capacities were allocated almost evenly across all available nodes. This spatial distribution is the direct result of fierce oligopolistic competition for highly lucrative market opportunities. Because participation in the system-wide markets – especially the reserve market – is overwhelmingly profitable, every single node represents a valuable access point to generate revenue. Consequently, to maximize their individual market shares and overall profits, the investors aggressively compete for, and fully exhaust, the available grid connection limits at every node across the network.

Regarding the **sizing (capacity)** of BESS, the framework revealed a fundamental divergence between power capacity (MW) and energy capacity (MWh). The optimal power capacity is primarily dictated by the competitive environment. Under the continuous

5 Synthesis of the Results

Cournot Inverse Demand mechanism, the investors formed a balanced oligopoly, sharing the power capacity provision equally to keep endogenous reserve prices high and avoid market cannibalization.

In contrast, the optimal energy capacity (the E/P ratio) is heavily dictated by the individual economic constraints of the investors – specifically, their cost of capital (WACC) – and the overwhelming profitability of the reserve market. The model proved that “patient” investors with low capital costs (e.g., 8%) deploy massive energy reservoirs (E/P ratio > 21) to compensate for a severe power bottleneck. Because the expensive inverter capacity (MW) is almost entirely blocked for reserve provision, investors oversize their cheap energy capacity (MWh) to still extract marginal revenues out of the spot market with the remaining fractional power.

Ultimately, the inclusion of both spot and reserve markets into the modeling framework uncovered a severe cross-market spillover effect. The results answer the research question by highlighting that optimal BESS allocation in a multi-service environment is not solely a function of geographic arbitrage. Instead, it is governed by an opportunity-cost logic: the strategic lock-up of power capacity for highly lucrative reserve markets forces the storage units into inefficient, slow-cycling, and occasionally loss-making operations on the spot market. Consequently, the true value of a BESS location and size is fundamentally shaped by the specific regulatory and economic design of the ancillary service markets it participates in.

5.2 Strengths and Limitations of the EPEC Methodology

The application of the EPEC framework to model the strategic allocation of BESS in a multi-service market proved to be both a powerful analytical tool and a significant mathematical challenge. By critically evaluating the modeling process, several distinct methodological strengths and necessary computational compromises can be identified.

5.2.1 Strengths of the Applied Framework

The primary strength of the EPEC methodology lies in its ability to simulate true oligopolistic competition through endogenous price formation. Unlike standard cost-minimization models where market prices are either exogenous inputs or remain independent of a single actor’s investment scale, the EPEC actively internalizes the market’s price response to newly installed BESS capacities.

During the development, the implementation of the *Residual Demand* formulation proved to be a critical methodological strength. By allowing investors to strictly optimize against the pre-calculated residual load of their rivals, the framework successfully eliminated the “cobweb” oscillations typically observed in iterative equilibrium models. Furthermore, the integration of a continuous *Cournot Inverse Demand Curve* for the aFRR market allowed for a realistic, stable representation of cartel-like profit maximization. This continuous formulation successfully prevented the non-linear solver from exploiting mathematical KKT-loopholes (the “price-dictator” exploit), ensuring that the resulting equilibrium is rooted in sound economic theory rather than mathematical artifacts.

5.2.2 Limitations and Mathematical Compromises

Despite these strengths, the non-linear and non-convex nature of coupled KKT conditions required several structural compromises to maintain computational tractability within the applied interior-point solver (Ipopt):

- **Continuous Approximation of Pay-as-Bid Markets:** Real-world ancillary service markets, such as the Austrian APG aFRR market, utilize a Pay-as-Bid clearing mechanism and require discrete minimum bid sizes (e.g., integer MW steps) [17]. Implementing a true Pay-as-Bid auction would have transformed the model into a Mixed-Integer Non-Linear Program with Equilibrium Constraints (MI-NLP). Given the state-of-the-art of optimization solvers, solving an MI-NLP of this scale within an iterative diagonalization loop is computationally intractable. Consequently, the continuous Cournot formulation was a mathematically necessary macroeconomic approximation.
- **KKT Relaxation and Numerical Noise:** To navigate the severe mathematical singularities that occurred during extreme peak-price and zero-price (curtailment) events, the strict complementarity conditions of the lower level had to be significantly relaxed ($\epsilon = 10.0$). While this provided the solver with the necessary gradient tube to ensure algorithmic convergence, it introduced microscopic numerical noise (e.g., fractional generator dispatch despite 0 €/MWh spot prices).
- **Linearized Battery Degradation:** Actual lithium-ion battery degradation is a highly complex, non-linear process heavily dependent on the Depth of Discharge (DoD) and specific cycle profiles, typically evaluated using Rainflow-Counting algorithms [8]. Because such algorithmic evaluations are non-differentiable, they cannot be embedded into an NLP solver’s constraints. Therefore, degradation had to be abstracted as a constant linear marginal cost per MWh throughput, slightly simplifying the operational reality of BESS.

5 Synthesis of the Results

- **Perfect Foresight (Deterministic Approach):** The model assumes perfect foresight regarding renewable generation and load profiles. Introducing stochastic optimization to account for forecast errors and actual balancing energy activation would have caused an exponential explosion in computation time, rendering the EPEC unsolvable on standard hardware.
- **Exclusion of Grid Tariffs and Network Fees:** In real-world electricity markets, BESS assets are often subjected to grid fees, taxes, and levies during their charging and discharging cycles. Because a BESS acts as both a consumer and a generator, these fees essentially act as a financial penalty on energy throughput, significantly shrinking the available arbitrage margins [10]. The current framework assumes a frictionless market without these grid fees, which leads to a slight overestimation of the profitability of pure spot market arbitrage.
- **Omission of Operations and Maintenance (O&M) Costs:** While the model rigorously accounts for annualized capital expenditures (CAPEX) and cyclic battery degradation, fixed annual O&M costs (typically estimated at around 1% of CAPEX [8]) were deliberately excluded from the objective function.

5.3 Upscaling and Transferability of Results

The EPEC modeling framework developed in this thesis, calibrated on a 5-bus network over a representative 24-hour period, serves as a rigorous mathematical proof-of-concept. It successfully demonstrates the intricate strategic dynamics of BESS investments in multi-service markets. However, rather than serving as a real-time operational tool, the primary value of this framework lies in its potential for advanced academic research and macroeconomic forecasting. To utilize this framework for comprehensive forecasts of future energy systems, long-term scenario planning, and data-driven policy consulting, several dimensions of upscaling must be addressed in future research.

5.3.1 Temporal and Spatial Upscaling

Currently, the model's temporal horizon is restricted to a single, highly volatile representative day. While this accurately captures diurnal arbitrage and physical battery cycling, it inherently neglects seasonal variations. Renewable generation patterns – such as prolonged dunkelflauten versus extended summer photovoltaic peaks – fundamentally alter the long-term profitability and state-of-charge management of BESS. Future applications of this model should upscale the temporal resolution by clustering

5.3 Upscaling and Transferability of Results

historical data into seasonal representative days (e.g., distinct profiles for spring, summer, autumn, and winter) or utilizing a rolling-horizon approach to approximate an 8760-hour year.

Spatially, the transferability of the model to large-scale, real-world power systems (such as the IEEE 39-bus system or the complete Austrian high-voltage transmission grid) is currently limited by the computational tractability of EPECs. As the number of nodes, lines, and strategic actors increases, the dimensionality of the coupled KKT conditions grows exponentially. Upscaling this framework to national grid levels will likely require the integration of advanced decomposition algorithms alongside the currently implemented diagonalization heuristic.

5.3.2 Transferability to European Market Designs

A critical aspect of this framework's transferability is its foundational market design. The implemented DC-OPF market clearing natively generates Locational Marginal Prices (LMPs), reflecting a *Nodal Pricing* system. While this is the standard market design in various U.S. markets, the European electricity market currently operates on a zonal pricing model ("copper plate" assumption), where internal grid congestion is resolved post-market via costly redispatch measures.

Therefore, the results of this thesis offer a powerful, forward-looking perspective. They demonstrate how strategic BESS investments would organically distribute themselves across the grid to relieve structural bottlenecks *if* a Nodal Pricing system were implemented. The framework can be highly valuable for the debate of transitioning towards smaller bidding zones or full nodal pricing, as it quantifies the precise economic incentives required to stimulate grid-supportive BESS allocation without relying on central-planner redispatch interventions.

5.3.3 Algorithmic Evolution towards MIPECs

Finally, the transferability of the operational results to real-world ancillary service markets is bounded by the continuous relaxation of the bidding variables. Real reserve markets demand discrete bid sizes (e.g., 1 MW increments). While the continuous *Cournot Inverse Demand* approximation utilized in this thesis is currently the most robust macroeconomic workaround, the ultimate goal for future research is the formulation of a Mixed-Integer Program with Equilibrium Constraints (MIPEC). Once commercial MI-NLP (Mixed-Integer Non-Linear Programming) solvers reach a maturity level capable of handling massive, coupled complementarity conditions, the framework can

5 Synthesis of the Results

be upgraded to reflect exact, discrete market auction rules without succumbing to the mathematical exploits currently observed in non-linear environments.

5.3.4 Transferability to Co-located Hybrid Configurations

The current framework exclusively simulates standalone BESS, where each unit interacts directly with the high-voltage transmission grid. However, in modern real-world project development, BESS units are increasingly co-located with renewable energy generators (such as solar PV and wind) in hybrid park configurations, sharing a single grid connection point. Transferring the EPEC framework to model such topologies represents a highly relevant future upscaling step. In a hybrid setup, the BESS can charge directly from local renewable surplus during high-yield periods without utilizing the public grid, thereby entirely avoiding transmission grid fees. Integrating these behind-the-meter dynamics would add a critical layer of strategic complexity, as investors would need to co-optimize grid-fee-free local charging against system-wide market arbitrage and reserve provision.

5.4 Lessons Learned from EPEC Modeling

The development and deployment of the EPEC framework for this thesis provided profound insights not only into BESS allocation but also into the fundamental mechanics of advanced energy system modeling. Reflecting on the iterative development process, several critical lessons learned emerged that can serve as best practices for future research in this highly complex field.

5.4.1 Algorithmic Limitations in Capturing Economic Rationality

A primary realization was that non-linear optimization solvers (such as Ipopt) are strictly mathematical engines that do not inherently account for economic rationale. If a mathematical loophole exists, the solver will exploit it to maximize the objective function, often leading to market-distorting paradoxes. This was profoundly evident in two instances during model development: First, the Zero-Price exploit in the spot market, where MPEC investors simulated microscopic, fictitious renewable curtailment using KKT slack tolerances solely to force the nodal market price artificially to 0 €/MWh, allowing them to charge their batteries for free. Second, the Price-Dictator phenomenon in the reserve market, where investors supplied near-zero reserve capacity but utilized the KKT complementarity slack to push the clearing price to the absolute penalty cap

(3,000 €/MW). These findings demonstrate that mathematical formulations in energy modeling should be closely guided by underlying economic principles. Replacing rigid KKT market clearing constraints with continuous economic mechanisms – such as the implemented Cournot Inverse Demand curve and penalty-driven soft constraints – is not a dilution of the model, but a mandatory requirement to ensure rational market behavior.

5.4.2 Challenges in Computing Simultaneous Market Equilibria

Modeling multiple strategic actors in a highly constrained power grid inherently invites the “cobweb” oscillations. Early model iterations revealed that when investors optimize based on the static, fixed decisions of their rivals, they continually evade each other in consecutive iterations, preventing the algorithm from converging to a Nash Equilibrium. The implementation of the *Residual Demand* concept – where an investor optimizes against the already shifted residual load profile of the entire system rather than the isolated actions of peers – was a fundamental breakthrough. It proved that in EPEC modeling, agents must react to the *net market state* rather than individual opponent variables to achieve algorithmic stability.

Finally, the development process highlighted that insisting on absolute mathematical purity often leads to computationally intractable models. In highly volatile power systems characterized by steep peak-load prices and zero-price renewable events, strict KKT complementarity conditions create severe mathematical singularities that crash interior-point solvers. Embracing pragmatic engineering solutions – such as relaxing the KKT conditions (e.g., $\epsilon = 10.0$), utilizing algorithmic damping factors ($\alpha = 0.25$) to suppress oscillation, and applying ex-post filters to remove microscopic numerical noise – proved absolutely essential. These compromises are not methodological flaws, but rather the necessary state-of-the-art tools required to make the theoretical elegance of EPECs practically applicable.

6 Conclusions and Outlook

This final chapter evaluates the overarching implications of the research conducted in this thesis. Rather than summarizing the quantitative results, it extracts the core lessons learned and distills the complex strategic interactions into a clear, fundamental takeaway.

The primary methodological achievement of this thesis is the successful development and stabilization of an Equilibrium Problem with Equilibrium Constraints (EPEC) tailored to Battery Energy Storage Systems (BESS) allocation. The results clearly demonstrate that the framework works: It successfully shifts the modeling paradigm from a naive central-planner cost-minimization approach to a decentralized, game-theoretic oligopoly. By reaching a stable Nash Equilibrium, the model proves that capturing the strategic self-interest of independent, profit-maximizing investors is both computationally feasible and absolutely essential for realistic energy system modeling.

However, the application of this framework revealed several unexpected realities and necessary compromises that fundamentally challenge traditional BESS allocation modeling:

- **Aggressive Siting:** Investors are driven by the profitability of system-wide markets and therefore aggressively distribute their assets across the entire grid to exhaust all available nodal connection limits and maximize their individual market shares.
- **Arbitrage-Reserve Trade-off:** The most significant operational finding is that lucrative capacity premiums in the aFRR market completely override traditional day-ahead logic. Investors lock up their expensive inverter capacity (MW) for reserve provision, creating a severe power bottleneck. To still participate in the spot market, they massively oversize their cheap energy capacity (MWh).
- **Methodological Pragmatism:** The continuous, non-linear nature of EPECs required significant engineering pragmatism. To prevent interior-point solvers from crashing at mathematical singularities or falling into oscillatory traps, relaxing KKT complementarity conditions and implementing a residual demand logic were mandatory compromises.
- **Future Evolution:** To overcome these continuous approximations, future research must aim towards Mixed-Integer Programs with Equilibrium Constraints (MIPEC). This would allow for the integration of discrete bidding steps, true Pay-as-Bid

6 Conclusions and Outlook

clearing, and non-differentiable battery degradation algorithms (e.g., Rainflow-Counting).

Ultimately, the most profound insight of this thesis extends beyond the mathematical equations. If one were to summarize the findings of this research in a single takeaway, it is this: *Optimal battery sizing and operation are not dictated by the physical availability of renewable energy, but by the regulatory design of the markets they participate in.*

When ancillary service markets are highly lucrative, strategic investors will act as rational opportunists. They will willingly accept inefficient, slow-cycling, and even loss-making operations on the spot market, simply because securing a dominant market share in the reserve market compensates for these opportunity costs. Therefore, evaluating battery investments based on single-market business cases or isolated grid metrics is an insufficient approach. To understand the true dimensioning and operational logic of storage systems in modern power grids, it is absolutely essential to account for the severe cross-market spillover effects driven by strategic investor behavior.

Bibliography

- [1] Hampfrey Zurua Otero, Cyrus W. Wekesa, and George Kimani Irungu. "Comprehensive Review of Energy Storage Technologies: Types, Applications, Optimal Sizing and Siting in Power Systems." In: *2022 IEEE PES/IAS PowerAfrica*. Aug. 2022, pp. 1–5. DOI: 10.1109/PowerAfrica53997.2022.9905263. URL: <https://ieeexplore.ieee.org/document/9905263> (visited on 06/02/2025) (cit. on p. 1).
- [2] Antonio J. Conejo and Carlos Ruiz. "Complementarity, Not Optimization, is the Language of Markets." In: *IEEE Open Access Journal of Power and Energy* 7 (2020), pp. 344–353. ISSN: 2687-7910. DOI: 10.1109/OAJPE.2020.3029134. URL: <https://ieeexplore.ieee.org/document/9215011> (visited on 02/22/2026) (cit. on pp. 2, 4, 5, 7, 9, 15).
- [3] Sheng Chen et al. "Equilibria in Electricity and Natural Gas Markets With Strategic Offers and Bids." In: *IEEE Transactions on Power Systems* 35.3 (May 2020), pp. 1956–1966. ISSN: 1558-0679. DOI: 10.1109/TPWRS.2019.2947646. URL: <https://ieeexplore.ieee.org/document/8869782> (visited on 02/22/2026) (cit. on pp. 2, 5, 17).
- [4] Michael L. Bynum et al. *Pyomo – Optimization Modeling in Python*. Third. Vol. 67. Springer Optimization and Its Applications. Springer, 2021. DOI: 10.1007/978-3-030-68928-5 (cit. on p. 2).
- [5] Andreas Wächter and Lorenz T. Biegler. "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming." In: *Mathematical Programming* 106.1 (2006), pp. 25–57. DOI: 10.1007/s10107-004-0559-y (cit. on p. 2).
- [6] Ahmad K. AL Ahmad and Reza Sirjani. "Optimal Allocation of Energy Storage System in Transmission System Considering Wind Power." In: *2020 7th International Conference on Electrical and Electronics Engineering (ICEEE)*. Apr. 2020, pp. 181–187. DOI: 10.1109/ICEEE49618.2020.9102603. URL: <https://ieeexplore.ieee.org/abstract/document/9102603> (visited on 06/05/2025) (cit. on p. 4).

Bibliography

- [7] Choton K. Das et al. "Optimal sizing of a utility-scale energy storage system in transmission networks to improve frequency response." In: *Journal of Energy Storage* 29 (June 2020), p. 101315. ISSN: 2352-152X. DOI: 10.1016/j.est.2020.101315. URL: <https://www.sciencedirect.com/science/article/pii/S2352152X19315646> (visited on 06/02/2025) (cit. on pp. 3, 4, 20, 22).
- [8] Hunyoung Shin and Jin Hur. "Optimal Energy Storage Sizing With Battery Augmentation for Renewable-Plus-Storage Power Plants." In: *IEEE Access* 8 (2020), pp. 187730–187743. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2020.3031197. URL: <https://ieeexplore.ieee.org/document/9223659/> (visited on 06/02/2025) (cit. on pp. 3, 4, 13, 22, 37, 38).
- [9] Hui Wang et al. "The Optimal Allocation and Operation of an Energy Storage System with High Penetration Grid-Connected Photovoltaic Systems." en. In: *Sustainability* 12.15 (Jan. 2020). Number: 15, p. 6154. ISSN: 2071-1050. DOI: 10.3390/su12156154. URL: <https://www.mdpi.com/2071-1050/12/15/6154> (visited on 06/05/2025) (cit. on p. 4).
- [10] Zeenat Hameed et al. "A business-oriented approach for battery energy storage placement in power systems." en. In: *Applied Energy* 298 (Sept. 2021), p. 117186. ISSN: 03062619. DOI: 10.1016/j.apenergy.2021.117186. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0306261921006140> (visited on 06/02/2025) (cit. on pp. 4, 38).
- [11] Farihan Mohamad, Jiashen Teh, and Ching-Ming Lai. "Optimum allocation of battery energy storage systems for power grid enhanced with solar energy." In: *Energy* 223 (May 2021), p. 120105. ISSN: 0360-5442. DOI: 10.1016/j.energy.2021.120105. URL: <https://www.sciencedirect.com/science/article/pii/S0360544221003546> (visited on 06/05/2025) (cit. on p. 4).
- [12] Hampfrey Odero, Cyrus Wekesa, and George Irungu. "Wind Energy Resource Prediction and Optimal Storage Sizing to Guarantee Dispatchability: A Case Study in the Kenyan Power Grid." en. In: *Journal of Electrical and Computer Engineering* 2022.1 (2022). eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1155/2022/4044757>, p. 4044757. ISSN: 2090-0155. DOI: 10.1155/2022/4044757. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2022/4044757> (visited on 06/02/2025) (cit. on p. 4).
- [13] Jinhu Wang et al. "Energy Storage Location and Capacity Planning Based on Distribution Network Partition and ResNet DNN." In: *2022 5th International Conference on Power and Energy Applications (ICPEA)*. Nov. 2022, pp. 388–392. DOI: 10.1109/ICPEA56363.2022.10052338. URL: <https://ieeexplore.ieee.org/document/10052338/> (visited on 06/02/2025) (cit. on pp. 4, 20).

- [14] Dominique Alonso Sørensen, Daniel Vázquez Pombo, and Esther Torres Iglesias. “Energy storage sizing for virtual inertia contribution based on ROCOF and local frequency dynamics.” en. In: *Energy Strategy Reviews* 47 (May 2023), p. 101094. ISSN: 2211467X. DOI: 10.1016/j.esr.2023.101094. URL: <https://linkinghub.elsevier.com/retrieve/pii/S2211467X23000445> (visited on 06/02/2025) (cit. on pp. 3, 4).
- [15] Berit Hanna Czock, Amelie Sitzmann, and Jonas Zinke. *The place beyond the lines - Efficient storage allocation in a spatially unbalanced power system with a high share of renewables*. eng. Working Paper 23/01. EWI Working Paper, 2023. URL: <https://www.econstor.eu/handle/10419/286374> (visited on 06/05/2025) (cit. on pp. 3, 4, 21).
- [16] Sonja Wogrin, Julián Barquín, and Efraim Centeno. “Capacity Expansion Equilibria in Liberalized Electricity Markets: An EPEC Approach.” In: *IEEE Transactions on Power Systems* 28.2 (May 2013), pp. 1531–1539. ISSN: 1558-0679. DOI: 10.1109/TPWRS.2012.2217510. URL: <https://ieeexplore.ieee.org/document/6331593> (visited on 10/13/2025) (cit. on pp. 5, 15–17).
- [17] Austrian Power Grid. *Secondary control (aFRR)*. Austrian Power Grid Market Platform. 2026. URL: <https://markt.apg.at/en/power-grid/balancing/secondary-control/> (visited on 02/23/2026) (cit. on pp. 13, 15, 37).
- [18] Mel T. Devine and Sauleh Siddiqui. “Strategic investment decisions in an oligopoly with a competitive fringe: An equilibrium problem with equilibrium constraints approach.” In: *European Journal of Operational Research* 306.3 (May 2023), pp. 1473–1494. ISSN: 0377-2217. DOI: 10.1016/j.ejor.2022.07.034. URL: <https://www.sciencedirect.com/science/article/pii/S0377221722005860> (visited on 10/13/2025) (cit. on pp. 15–17, 20).
- [19] HSL. *A collection of Fortran codes for large scale scientific computation*. See <https://licences.stfc.ac.uk/product/coin-hsl-archive> for the Coin-HSL Archive version. 2024. URL: <http://www.hsl.rl.ac.uk/> (cit. on p. 17).
- [20] Austrian Power Grid. *Actual Generation per Production Type (Data for 2025-09-11)*. Accessed via APG Market Transparency Platform; Data Export for September 11, 2025. 2025. URL: <https://markt.apg.at/en/transparency/generation/generation-per-production-type/> (visited on 02/28/2026) (cit. on p. 21).